

GWANGJU, Korea, Republic of

WMO: 471560

Lat: 35.1731N

Lon: 126.8917E

Elev: 242

StdP: 14.57

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.1	22.9	4.0	6.8	30.2	7.4	8.1	30.9	16.1	36.0	13.7	33.6	5.2	50	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.8	91.4	77.8	89.2	76.8	87.1	75.6	80.4	86.4	79.1	85.5	78.0	84.4	6.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.9	151.4	83.4	77.4	143.7	82.3	76.2	137.8	81.9	44.3	86.3	42.9	86.0	41.7	84.6	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.6	13.1	11.3	15.9	94.7	3.5	2.4	13.3	96.4	11.3	97.8	9.3	99.2	6.8	101.0
			13.6	81.0	3.1	2.3	11.4	82.7	9.6	84.1	7.8	85.4	5.6	87.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	58.2	34.2	37.8	46.1	56.7	66.2	73.5	79.4	80.3	72.5	61.9	49.9	38.3	(6)
(7)		DBStd	16.93	6.09	6.67	6.65	6.22	4.67	3.42	4.39	4.26	4.68	5.40	6.84	6.42	(7)
(8)		HDD50	1461	491	344	158	14	0	0	0	0	0	2	86	366	(8)
(9)		HDD65	4009	956	761	587	255	42	0	0	0	3	123	454	828	(9)
(10)		CDD50	4447	1	4	36	216	501	706	911	940	676	371	83	2	(10)
(11)		CDD65	1520	0	0	0	7	78	257	446	475	228	28	1	0	(11)
(12)		CDH74	12220	0	0	0	57	514	1462	4060	4706	1324	95	2	0	(12)
(13)		CDH80	3717	0	0	0	2	81	328	1336	1692	275	3	0	0	(13)
(14)	Wind	WSAvg	4.5	4.2	4.4	4.8	5.0	4.8	4.6	5.6	4.9	4.3	3.6	3.7	3.9	(14)
(15)	Precipitation	PrecAvg	52.6	1.4	1.7	2.6	3.9	3.9	7.1	10.7	10.0	5.9	2.2	2.2	1.3	(15)
(16)		PrecMax	79.5	5.0	4.3	6.7	8.5	8.9	21.4	27.3	21.2	18.7	6.8	6.9	2.9	(16)
(17)		PrecMin	30.1	0.2	0.1	0.4	0.6	0.2	1.0	1.4	1.1	0.4	0.1	0.6	0.1	(17)
(18)		PrecStd	13.4	1.0	1.1	1.6	2.2	2.2	4.9	5.7	5.1	4.3	1.7	1.5	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.0	62.4	69.6	79.3	85.7	88.7	94.6	94.7	88.7	79.7	71.5	59.2	(19)
(20)		0.4%	MCWB	48.8	54.7	54.0	61.5	64.1	69.1	78.8	78.9	74.0	65.1	61.3	51.2	(20)
(21)		2%	DB	49.8	56.4	64.8	75.2	82.1	85.8	91.4	92.0	85.5	76.6	67.1	54.6	(21)
(22)		2%	MCWB	42.5	48.2	51.4	59.3	63.0	68.9	78.3	78.2	72.5	63.0	57.6	47.0	(22)
(23)		5%	DB	46.4	52.6	61.2	71.9	79.4	83.8	89.0	90.0	83.1	74.1	63.9	51.4	(23)
(24)		5%	MCWB	40.2	44.5	50.5	57.0	62.3	68.8	77.4	77.7	71.2	61.9	55.6	45.3	(24)
(25)		10%	DB	43.5	48.9	57.9	68.4	76.7	81.6	86.7	87.8	80.9	71.4	61.0	48.4	(25)
(26)		10%	MCWB	37.9	41.4	47.9	55.3	61.6	68.3	76.8	76.9	70.1	60.9	53.4	42.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.4	57.7	59.0	67.0	71.5	76.1	82.2	82.1	77.5	71.8	64.0	53.7	(27)
(28)		0.4%	MCDWB	53.0	60.2	63.0	72.9	77.4	80.5	87.2	88.4	82.7	75.4	68.5	57.5	(28)
(29)		2%	WB	44.5	50.3	55.7	63.0	68.1	74.3	80.6	80.6	76.1	67.9	60.9	49.1	(29)
(30)		2%	MCDWB	48.5	54.3	60.6	69.9	73.3	78.8	86.3	86.9	81.8	71.1	64.1	53.2	(30)
(31)		5%	WB	41.3	45.4	52.3	60.3	66.0	72.9	79.2	79.4	74.6	64.9	57.5	45.8	(31)
(32)		5%	MCDWB	45.4	50.9	58.8	67.2	74.0	78.0	85.4	85.9	80.2	70.0	62.4	50.7	(32)
(33)		10%	WB	38.3	42.4	49.2	58.0	64.3	71.4	78.0	78.4	72.8	62.6	54.7	43.1	(33)
(34)		10%	MCDWB	42.5	48.1	56.5	64.8	72.7	77.1	84.4	85.0	78.0	69.0	59.8	48.0	(34)
(35)	Mean Daily Temperature Range		MDBR	12.8	15.2	17.6	19.1	18.2	14.4	11.0	11.8	14.0	16.6	15.5	13.4	(35)
(36)		5% DB	MCDBR	16.7	20.5	23.0	24.5	23.4	19.1	14.3	14.8	16.4	19.6	19.3	18.3	(36)
(37)			MCWBR	10.8	12.6	12.1	10.7	8.8	6.4	4.5	4.5	5.8	7.9	10.4	12.0	(37)
(38)			MCDBR	14.9	17.1	18.5	17.4	16.6	11.9	12.4	12.5	12.3	13.3	14.7	16.2	(38)
(39)		5% WB	MCWBR	11.2	13.0	12.6	10.5	8.7	5.7	4.8	4.7	6.2	8.3	10.5	12.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.446	0.486	0.558	0.576	0.592	0.655	0.610	0.561	0.497	0.470	0.456	0.445	(40)	
(41)		taud	1.936	1.876	1.731	1.740	1.751	1.693	1.847	1.970	2.082	2.021	1.988	1.949	(41)	
(42)		Ebn at Noon	225	232	227	232	230	216	225	233	242	235	220	215	(42)	
(43)		Edh at Noon	46	55	69	72	72	76	65	56	48	47	44	43	(43)	
(44)	All-Sky Solar Radiation	RadAvg	775	1026	1383	1598	1747	1557	1427	1468	1316	1171	829	686	(44)	
(45)		RadStd	60	90	123	128	187	143	197	203	132	102	89	47	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	-1.58	N/A	N/A	N/A	N/A	N/A	+176	+112	(46)
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+115	N/A	(47)

Nomenclature: See separate page